Collaborative Classification of Hyperspectral and LiDAR Data Based on Dynamic Multiple Fractional Fourier Domains Fusion

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***Abstract*— Collaboratively utilizing the complementary information provided by hyperspectral imagery and light detection and ranging (LiDAR) data will extend the applications associated with land cover recognition and mapping. Existing joint classification algorithms mainly focus on learning complementary patterns in the pure spatial domain, while paying little attention to complementary cues in the spatial–frequency domain. The model's expressive capability of these methods may be limited by an upper bound subject to the spatial domain. To fill this gap, a dynamic multiple fractional Fourier domains fusion (DMFraF) is proposed for the joint classification of hyperspectral and LiDAR data. First, to comprehensively learn the complementary patterns between hyperspectral image (HSI) and LiDAR data, we transform the features of two modalities into multiple fractional domains containing different spatial–frequency components for multimodal fusion. Second, to obtain the optimal representation from the multimodal features of multiple fractional domains, we propose a dynamic fusion scheme guided by the optimal transport (OT) technique, which can dynamically adjust the contributions from different fractional domains. Finally, to extract purer modality-specific features, we propose a channel aggregation Transformer encoder with channel aggregation Transformer (C^2AT encoder), to aggregate channel-wise features of central pixels into the spatial branch and compress interference from noisy surroundings. Extensive experiments and analysis on three hyperspectral and LiDAR datasets suggest the superiority of the proposed method.**

***Index Terms*— Collaborative classification, fractional Fourier transform (FrFT), hyperspectral image (HSI), light detection and ranging (LiDAR), optimal transport (OT).**

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I. Introduction

THE rapid advancement of remote sensing (RS) imaging technology has ushered in new possibilities for a better understanding of intelligent Earth observation [1], [2], [3]. Fueled by the increasing availability of multisource and multimodal RS data, a wide variety of RS applications have emerged, such as ground object classification [4], [5], scene understanding [6], [7], mineral exploration and mapping [8], [9], and change monitoring [10], [11]. Hyperspectral images (HSIs), as one of the most concerning types of RS data, benefits from rich spectral information, providing a natural advantage in the mission of ground material identification [12], [13], [14]. To fully exploit the spatial and spectral advantages of HSIs, various effective models [15], [16], [17] and learning schemes [18], [19], [20], [21] are proposed for feature extraction and data analysis of HSIs. For instance, Li et al. [16] proposed a dual-branch architecture to separately learn spatial and spectral information, while developing two distinct attention mechanisms to, respectively, enhance spatial and spectral feature representations. Sun et al. [17] innovatively modified the Transformer network by introducing a novel Gaussian-weighted feature tokenization approach for effective spatial–spectral semantic information encoding. Furthermore, recent methodological advancements have begun challenging the independent and identically distributed (i.i.d.) assumption, addressing more complex and challenging HSIC tasks. For example, Qin et al. [18] proposed a feature-disentanglement-based cross-domain few-shot learning framework, which mitigates model bias induced by extensive source domain training data through compression of domain-specific features. Concurrently, Xie et al. [20] developed a feature-consistency-guided prototype learning mechanism to reduce open-set risk, enabling robust model performance under open-set conditions.

While HSIs provide rich spectral information for land-cover identification, it lacks the information of object height. Relying solely on such a passive hyperspectral sensing mode is still susceptible to the influence of volatile imaging conditions. In the presence of objects exhibiting similar spectral characteristics but differing elevations, it is significantly challenging to identify the target only based on the spectral–spatial pattern provided by the HSI. Fortunately, by collaboratively utilizing the height information contained in light detection and ranging (LiDAR) data, these pixels with indiscernible spectral–spatial

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patterns but different semantics can get a chance to be distinguished [22], [23]. In this context, recent years have witnessed a surge of interest [24], [25] in integrating HSI and LiDAR data, driven by the impressive success of multimodal models in machine learning. Ideally, the joint land-cover classification using complementary cues between HSI and LiDAR data can achieve superior performance compared to either of them in isolation. However, confounding effects and dependencies among such heterogeneous multimodal data sources inevitably introduce unwanted variability and bias, posing a significant challenge for further breaking of performance bottlenecks [26], [27]. Moreover, due to extraneous variables and multiple distributions in HSI and LiDAR data, it can be a laborious task to accurately uncover the actual relationship between variables for a given downstream task. Therefore, the core challenge in joint classification tasks is that how to address the side effects of confounders and underlying dependencies among different data sources.

Owing to the powerful data fitting and representative ability, many practitioners resort to deep neural networks (DNNs) [28], [29]. The majority of early collaborative classification methodologies of HSI and LiDAR data were primarily based on convolutional neural network (CNN) architectures. They typically obey a bi-level design rule, i.e., a well-designed private encoder for each modality based on modality-specific properties, and a parameter-shared fusion network that aims to capture complementary patterns between modalities. For instance, some methods [30], [31] focused on designing effective CNN-based individual encoders to capture the spectral–spatial information of HSIs and the elevation features from LiDAR, while other methods [32], [33], [34] concentrated on developing better algorithms to integrate the heterogeneous modal features of HSI and LiDAR data.

Specifically, Hang et al. [30] employed a coupled CNN architecture (CoupledCNN) to separately extract hyperspectral and LiDAR features, subsequently implementing feature-level and decision-level fusion strategies for joint classification. Roy et al. [31] proposed a morphological feature learning framework that enhances hyperspectral-LiDAR joint classification through modality-specific morphological feature extraction. To achieve superior multimodal feature integration, Zhang et al. [32] introduced a dynamic structure optimization method that systematically leverages complementary advantages between the two modalities. In a groundbreaking approach, Yu et al. [34] pioneered the exploration of cross-modal relationships between hyperspectral and LiDAR data from an intrinsic image decomposition perspective, enabling comprehensive fusion of their respective modality-specific advantages.

Thanks to the advantages of global interaction and sequential processing, Transformers have shown great potential in various vision tasks [35], [36], [37]. In addition, the Transformer's sequential procedure provides a unique advantage for handling multimodal data, and its token mixers have been instrumental in enhancing perception across different modalities [38], [39]. By employing a private yet simple data-to-sequence tokenization for each modality, it becomes convenient to encode data from different modalities and bridging of complex representations through a shared parameter space (i.e., token mixer).

For example, Xue et al. [40] proposed a deep hierarchical vision Transformer with three branches designed to integrate spatial representations from HSI, spectral representations from HSI, and height information from LiDAR data. Yao et al. [41] achieved shared projection for heterogeneous modal data through separated convolutions, and captured complementary clues between modalities using an extended vision Transformer (EXViT) based on attention fusion. Roy et al. [42] proposed a multimodal fusion Transformer (MFT) to address the issue of extreme information imbalance between modalities by incorporating LiDAR representations, which can have much lower information content compared to hyperspectral data, as an external embedding similar to a class token in the token mixer for modalities fusion.

Beyond architectural innovations, optimization-driven methodologies for multisource classification have emerged as a promising research direction. Kong et al. [43] implemented cross-modal data integration through a spectral prompting mechanism that adaptively aligns heterogeneous data representations. Inspired by continual learning paradigms, Wang et al. [29] proposed a representation-enhanced status replay network to mitigate intermodal representation divergence during incremental fusion processes. In parallel, Ding et al. [44] devised an uncertainty-aware contrastive learning framework that enables semi-supervised multisource classification through probabilistic estimation of feature reliability. Feng et al. [45] developed a gradient consistency optimization paradigm to reduce intermodal feature discrepancies, effectively eliminating the requirement for explicit intermodal interaction mechanisms. Meanwhile, some works attempt to extend multisource classification tasks to non-i.i.d. scenarios. For example, Han et al. [46] proposed a novel data-aware and model-aware distribution enhancement methodology specifically designed to address cross-scene multisource classification challenges. Gao et al. [47] enhanced model robustness through a sharpness-aware minimization loss optimization framework, enabling effective generalization to novel joint classification scenarios. In contrast, Yang et al. [48] developed a relation-learning-based domain adaptation methodology to mitigate distributional discrepancies between source-domain and target-domain multimodal data.

Although these methods have achieved impressive results in the joint classification task of HSI and LiDAR data, they lack exploration of intramodality representations and intermodality complementarity in the frequency domain. These methods mainly focus on designing better feature interaction methods and typically explore complementary information between modalities only in the spatial domain. This will inevitably lead to the expressive capability of joint features being implicitly limited by an upper bound subject to the spatial domain. Although Zhao et al. [49] investigated multisource collaborative classification tasks from a frequency-domain perspective and proposed a fractional Fourier image Transformer (FrIT), their work primarily concentrated on frequency-domain feature extraction rather than multimodal feature interaction.

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Inspired by the demonstrated superiority of fractional Fourier transform (FrFT) in processing nonstationary signals [50], we aim to exploit the inherent capability of the fractional domain to encompass multidomain information characteristics. Multiple fractional domains will offer the further potential for exploring complementary cues between HSI and LiDAR data, thereby overcoming the representational limitations imposed by exclusive reliance on spatial-domain features. In particular, by setting different fractional orders, multiple fractional domain representations can be provided for the multimodal data, where each fractional order may reflect different data relationships. However, while multiple fractional domains offer enriched perspectives for multimodal fusion, the extraction of task-relevant information across these domains remains a persistent challenge. optimal transport (OT) theory provides a principled framework for measuring the minimum transportation cost between feature distributions, explicitly quantifying their inherent discrepancies. This distributional discrepancy inherently correlates with representational divergence between different features, thereby establishing a new possibility for quantifying intermodal representation differences within distinct fractional domains. Consequently, the development of a quantitative metric capable of evaluating task-specific contributions from distinct fractional domains would eliminate heuristic configurations of fractional orders while enabling synergistic integration of cross-domain multimodal features.

To overcome these limitations, a novel dynamic multiple fractional Fourier domains fusion (DMFraF) is proposed for the joint classification of hyperspectral and LiDAR data. As shown in Fig. 1(b), DMFraF aims to systematically explore intermodal complementary information across distinct fractional domains while harmonizing their task-specific relevance for classification objectives. First, to fully explore the complementary cues between hyperspectral and LiDAR modalities, modality-specific features of HSI and LiDAR data are transformed into multiple fractional domains with different orders for multimodal fusion. Then, to dynamically adjust the task-specific contributions of multimodal representations from different fractional domains, we propose a dynamic fusion scheme guided by the OT distance. By calculating the OT distances between the two modalities, the intermodality heterogeneity will be implicitly quantified. Finally, to extract modality-specific features, we specifically propose a channel aggregation Transformer encoder based on channel aggregation Transformer (C^2AT encoder) tailored to HSI and LiDAR data. The C^2AT effectively aggregates channel-wise features of central pixels into the spatial branch and compresses interference from noisy pixels.

In summary, the main contributions of this article can be summarized as follows.

1) We incorporate the FrFT into multimodal fusion and propose a multiple fractional Fourier domains fusion Network for the joint classification of HSI and LiDAR data. Multiple fractional domains characterized by distinct spatial and frequency components can provide geometrically diversified feature perspectives, facilitating better capture of complementary cues and task-discriminative information between HSI and LiDAR data.
2) A dynamic fusion scheme based on the OT technique is proposed to obtain the optimal fusion representation from multiple fractional domains. By estimating the modality heterogeneity across fractional domains using OT distance, DMFraF can dynamically adjust the contributions from different domains for each instance.
3) We designed a channel aggregation Transformer encoder based on central cross-attention to extract modality-specific features. By aggregating the channel-wise features of central pixels into the spatial branch, it can avoid interference from noisy surroundings during the learning of modality-specific attributes.

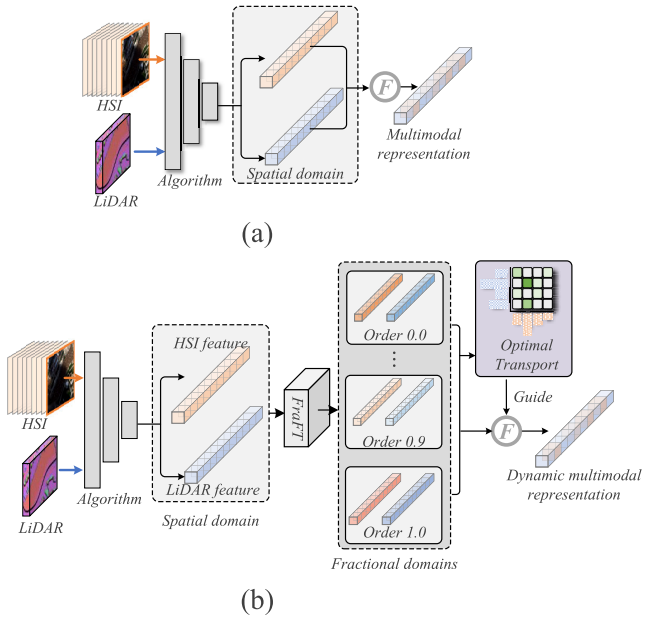


Fig. 1. Schematic workflow of our proposed method compared with existing joint classification approaches. (a) Existing joint classification algorithms mainly focus on modeling complementary patterns in the spatial-only domain. (b) Our proposed DMFraF can capture complementary cues between HSI and LiDAR data from multiple fractional domains, and dynamically adjust the contributions of multimodal representations from different fractional domains through the OT distance.

The remainder of this article is organized as follows. In Section II, the proposed DMFraF is technically presented in detail. Section III reports and analyzes experimental results. Finally, the conclusion is outlined in Section IV.

II. Methodology

In this section, the overall workflow of DMFraF is introduced first. Then, the detailed designs of three highlighted constituent modules in the DMFraF are presented.

A. Method Overview

The overall framework of the proposed DMFraF is illustrated in Fig. 2. For a given pair of hyperspectral and LiDAR data, we initially patch them, treating the hyperspectral patches as input instances for the model. The DMFraF mainly consists of three highlighted constituent modules, i.e., a channel aggregation Transformer encoder based on central cross-attention,

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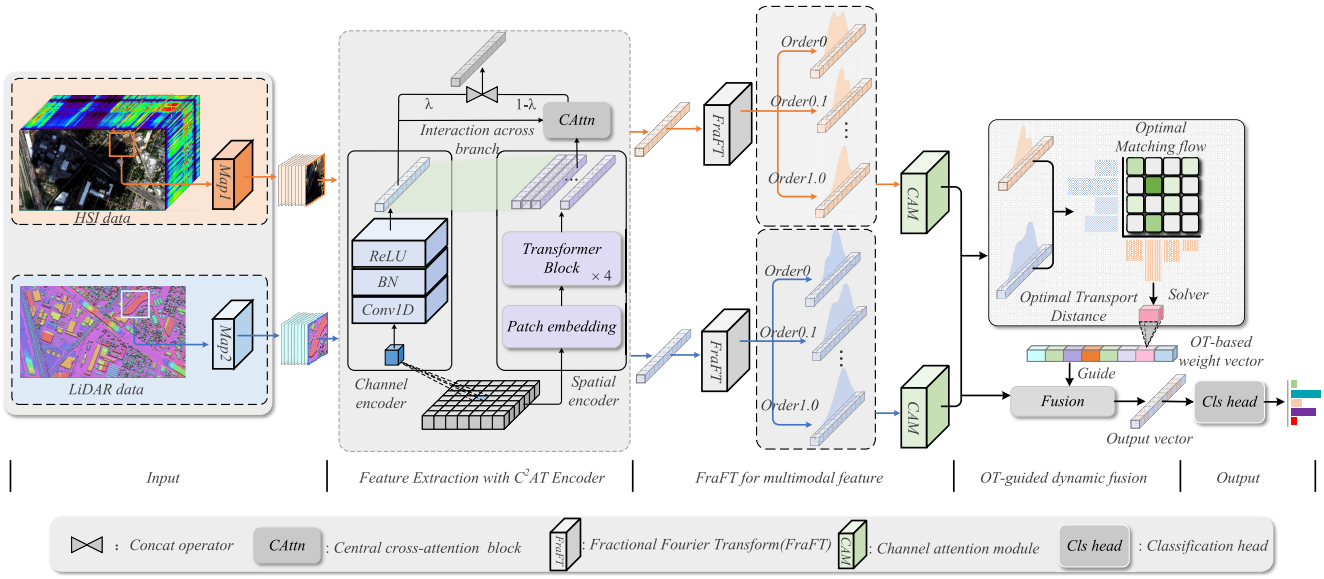


Fig. 2. Schematic workflow of the proposed DMFraF, which mainly consists of three highlighted constituent modules, i.e., feature extraction with C^{2}AT encoder, which is treated in Section II-B, FraFT for multimodal feature, which is treated in Section II-C, and OT-guided dynamic fusion, which is treated in Section II-D.

multiorder fractional Fourier domain representation for multimodal feature learning, and OT-guided fusion. The C^{2}AT aims to extract the spatial and channel features of HSI and LiDAR data, i.e., spatial and spectral features of HSI data, as well as spatial and channel features of mapped LiDAR data. Multiorder FraFT can transform feature into multiple fractional domains to better explore complementary cues between modalities. Finally, based on the OT technique, DMFraF can dynamically adjust the contributions of multimodal representations in different fractional domains according to the different input instance pairs.

B. Multimodal Feature Extraction With C^{2}at Encoder

Formally, denoted the HSI and LiDAR image, each collecting $H \times W$ pixels, as $\mathcal{H} \in \mathbb{R}^{H\times W\times B^h}$ and $\mathcal{I} \in \mathbb{R}^{H\times W\times B^\ell}$, where B^h and B^ℓ represent the channels of HSI and LiDAR image, respectively. The essence of multimodal joint classification lies in predicting the category of each pixel by incorporating data from two modalities, i.e., HSI and LiDAR, each containing C categories. In order to take advantage of the spatial pattern of two modalities, we partition the HSI and LiDAR images into pairwise input patches $\mathbf{X}^h = \{\mathbf{x}_i^h\}_i^N \in \mathbb{R}^{p\times p\times B^h}$, $\mathbf{X}^\ell = \{\mathbf{x}_i^\ell\}_i^N \in \mathbb{R}^{p\times p\times B^\ell}$, where the p is the token size and $N = H \times W$. In this section, a C^{2}AT encoder is designed to extract the spatial and channel information of HSI and LiDAR data.

As data points from the two modalities lie in distinct feature spaces, in order to facilitate the encoding of multimodal representations via a weight-shared model, two private mapping layers are initially employed to transform them into a common feature space. Considering the extremely imbalanced information between the HSI and LiDAR image, we leverage a simple identity mapping as the projection layer for the hyperspectral modality, while the projection layer for the LiDAR modality primarily relies on channel-wise repetition stacking. It is important to note that these two mapping layers will not change the spatial dimensions of the data. Defined the pairwise representations of the two modalities in the common space as $\{\tilde{\mathbf{X}}^h, \tilde{\mathbf{X}}^\ell\} \in \mathbb{R}^{p\times p\times B}$, where B represents the dimensions of the shared feature space and it is equal to B_h. Considering that the channel features and spatial features of different modalities should encompass representations from distinct views, the feature extraction for each modality is decoupled into two pipelines, namely, the spatial branch and the channel branch. Specifically, the Transformer is adopted as the learner of the spatial branch to capture long-range contextual relationships of the patches. With respect to the channel branch, we only take the central pixel as the input of the channel pipeline, enabling the model to effectively capture detailed interchannel correlations.

Mathematically, denoted the θ_{spa} and θ_{cha} as the network of spatial branch and channel branch, respectively. For the sake of symbol simplicity, we use h/ℓ to represent the application of the same operation on either the hyperspectral or LiDAR data flow. Particularly, the θ_{spa} consists of a patch embedding layer and four Transformer blocks based on self-attention, then the spatial feature can be obtained by

$$\boldsymbol{f}_{\text{spa}}^{h/\ell} = \theta_{\text{spa}}\left(\tilde{\mathbf{X}}^{h/\ell}\right) \in \mathbb{R}^{n_p\times B^{\text{emb}}} \tag{1}$$

where the $n_p = p/p^{\text{emb}}$ and p^{emb} is the patch size of the Transformer. Next, to extract interchannel correlations, we fed the common representation of central pixel $\tilde{\mathbf{X}}_c^{h/\ell} \in \mathbb{R}^{1\times 1\times B}$ separately into a 1-D convolutional block θ_{cha}, that is,

$$\boldsymbol{f}_{\text{cha}}^{h/\ell} = \theta_{\text{cha}}\left(\tilde{\mathbf{X}}_c^{h/\ell}\right) \in \mathbb{R}^{1\times B^{\text{emb}}} \tag{2}$$

where the θ_{cha} is composed of a 1-D convolutional layer, a batch normalization (BN) layer, and an ReLU layer.

After obtaining the spatial and channel feature representations, a central cross-attention (CAttn) block is designed to aggregate channel information into the spatial branch. Different from prior cross-attention mechanisms in HSI classification

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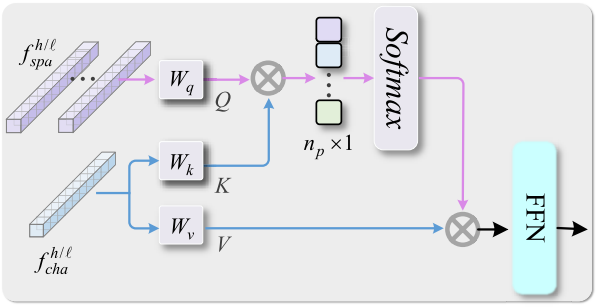


Fig. 3. Illustration of CAttn block. CAttn block aims to achieve the interaction between the central pixel embedding and spatial embeddings.

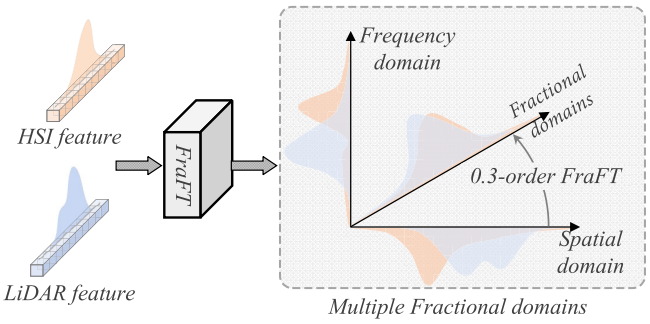


Fig. 4. Schematic of multiorder fractional Fourier transform.

(HSIC), only the central features are transformed as the key and value in the C^{2}AT encoder. As shown in Fig. 3, the interaction of channel and spatial pipelines can be expressed as

$$\begin{aligned} &Q = \boldsymbol{f}_{\text{spa}}^{h/\ell} W_q, \quad K = \boldsymbol{f}_{\text{cha}}^{h/\ell} W_k, \quad V = \boldsymbol{f}_{\text{cha}}^{h/\ell} W_v \\ &A = \text{softmax}\left(QK^T / \sqrt{B^{\text{emb}}} \right), \quad \boldsymbol{f}_{\text{agg}}^{h/\ell} = AV \end{aligned} \tag{3}$$

where $W_q, W_k, W_v \in \mathbb{R}^{B^{\text{emb}} \times B^{\text{emb}}}$ are weight matrices. This coupled representation $\boldsymbol{f}_{\text{agg}}^{h/\ell}$ not only reduces the heterogeneity between the features obtained from two heterogeneous architectures, but also encompasses more diverse representations. Notably, the core innovation of our C^{2}AT encoder lies in its decoupled learning paradigm for spatial and channel-wise information, coupled with attention-based feature aggregation. Therefore, the structural architecture of the CAttn block maintains operational parity with conventional cross-attention mechanisms, but they differ in their interaction targets. Specifically, the conventional cross-attention mechanism typically interacts with embeddings between spatial image patches, whereas the CAttn module strategically focuses on interactions between the central pixel embedding and its spatial branch embeddings. Finally, a concatenation operation based on balanced parameters λ is employed to fuse the features from the two branches, i.e., $\boldsymbol{f}^{h/\ell} = \text{Concat}(\lambda \boldsymbol{f}_{\text{cha}}^{h/\ell}, (1-\lambda)\boldsymbol{f}_{\text{agg}}^{h/\ell}) \in \mathbb{R}^{2B^{\text{emb}}}$. For simplicity, denoted $2B^{\text{emb}}$ as B^o, then the output features of HSI and LiDAR data is $\boldsymbol{f}^h, \boldsymbol{f}^\ell \in \mathbb{R}^{B^o}$.

C. Multiorder Fractional Domain Representation for Multimodal Feature Leaning

Conventional hyperspectral and LiDAR data classification methods primarily fuse $\boldsymbol{f}^h$ and $\boldsymbol{f}^\ell$ in the spatial domain. For nonstationary signals such as hyperspectral data, they will be prone to exhibit discriminative features in the fractional domain that includes joint spatial–frequency properties [50]. Therefore, explicitly exploring the complementarity between modalities by transforming features into multiple fractional domains will provide richer complementary representations.

Unlike the Fourier transform, which only considers attributes in the pure frequency domain, the FraFT can transform data into a broader range of fractional domain feature spaces by controlling the contributions between frequency and spatial attributes (i.e., shown in Fig. 4). We leverage discrete FrFT (DFrFT) to transform the multimodal feature $\boldsymbol{f}^{h/\ell}$ and viewed them as a sequence $\boldsymbol{f}(u)$. It is important to note that the FrFT method is flexible and can be chosen based on the specific data properties. In DMFraF, for the sake of implementation convenience, we followed FrIT [49] and selected the DFrFT. Notably, although both FrIT and ours utilize the FraFT, the core motivation of FrIT is to replace the token mixer (i.e., self-attention module) in the Transformer layer with the FraFT. In contrast, the core motivation of our method is to explore the complementary information between hyperspectral and LiDAR data across multiple fractional Fourier domains. Denoted that the transform angle $\alpha = \rho\pi/2$, where the ρ represent the order of FraFT, and the representation of fractional domain can be obtained by

$$\boldsymbol{f}^\rho(v) = \mathcal{F}^\rho\left[\boldsymbol{f}(u)\right]. \tag{4}$$

If the $\alpha \in \{2m\pi + (0, \pi)\}$ and m is the integer

$$\begin{aligned} \mathcal{F}^\rho\left[\boldsymbol{f}(u)\right] = &\sqrt{\frac{\sin\alpha - j\cos\alpha}{B^o}} \cdot \exp\left(\frac{j}{2}\cot\alpha v^2\right) \\ &\cdot \sum_{u=0}^{B^o-1} \exp\left(-j\frac{2\pi uv}{B^o}\right) \cdot \exp\left(\frac{j}{2}\cot\alpha u^2\right)\boldsymbol{f}(u) \end{aligned} \tag{5}$$

where the $0 \le v \le B^o - 1$. Correspondingly, when the $\alpha \in \{2m\pi + (-\pi, 0)\}$, it can be represented as

$$\begin{aligned} \mathcal{F}^\rho\left[\boldsymbol{f}(u)\right] = &\sqrt{\frac{-\sin\alpha + j\cos\alpha}{B^o}} \cdot \exp\left(\frac{j}{2}\cot\alpha v^2\right) \\ &\cdot \sum_{u=0}^{B^o-1} \exp\left(j\frac{2\pi uv}{B^o}\right) \cdot \exp\left(\frac{j}{2}\cot\alpha u^2\right)\boldsymbol{f}(u). \end{aligned} \tag{6}$$

It is worth noting that when $\rho = 0$, i.e., $\alpha = 0$, $\boldsymbol{f}(u) = \mathcal{F}^\rho[\boldsymbol{f}(u)]$, indicating that the features are in the original spatial domain. Inversely, when $\rho = 1$, the FrFT degenerates into the standard Fourier transform.

To explore the complementary cues between two modalities from multiple fractional domains, eleven fractional domains ranging from 0 to 1 are selected, i.e., $\rho \in [0 : 0.1 : 1]$. It should be noted that two endpoints are retained. Hence, the features of both the pure frequency domain and the spatial domain are preserved. As the original features space are within the spatial domain, i.e., $\rho = 0$, we directly connect the original features to the subsequent feature sets via a shortcut connection. Then

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the feature sets including multiple fractional domains for two modalities can be defined as $\boldsymbol{F}^h = \{\boldsymbol{f}^{h,0}, \boldsymbol{f}^{h,0.1}, \ldots, \boldsymbol{f}^{h,1.0}\}$ and $\boldsymbol{F}^\ell = \{\boldsymbol{f}^{\ell,0}, \boldsymbol{f}^{\ell,0.1}, \ldots, \boldsymbol{f}^{\ell,1.0}\}$. This setting allows the model to explore complementary information across modalities from nearly all fractional domains between the pure spatial and frequency domains, which avoids the heuristic setting of the fractional order.

After obtaining feature representations in multiple domains, we initially pass the feature sets of two modalities through a channel-wise attention module (CAM) [51]. This block aims to refine the intramodality representation of important fractional domains for different modalities by leveraging interchannel correlations. To be specific, the feature sets from each modality are first concatenated into a feature vector along the channel dimension. Subsequently, a 1-D convolution layer with 1×1 kernel size and 1 stride is applied, similar to the standard CAM, to learn channel-level weights. Finally, the feature vector is reshaped to match the dimensions of the original feature set.

D. OT-Guided Dynamic Fusion for Multimodal Feature

The feature sets from multiple fractional domains contain different multimodal representations, posing a new challenge of selecting optimal fractional domain features. Extending on this ideal, we hope to incorporate the information from each domain into multimodal fusion. OT [52], as a fundamental family of matching two probability distributions, is able to measure the optimal matching cost based on a structural measurement between two data geometric [53]. Moreover, as it allows to learn one-to-many stochastic mappings with tunable level of sample diversity, OT can provide a transport plan to transfer one distribution to another and is widely used in generative modeling or some ill-posed problems [54]. In DMFraF, OT is not leveraged as a regularizer to constrain feature representation with potential divergence like [32], as our core motivation is to extract complementary multimodal representations from multiple fractional domains and balance their contributions to the classification task. Therefore, to the above-mentioned aim, we dynamically construct unique multimodal representations for different input instances by measuring the intermodality heterogeneity across different fractional domains.

Specifically, by measuring the OT distance of hyperspectral and LiDAR modalities in different fractional domains, it can reflect the information density of complementary cues and be viewed as the guidance of the dynamic fusion for different inputs. Taking the pairwise multimodal embeddings $\{\boldsymbol{f}^{h,\rho}, \boldsymbol{f}^{\ell,\rho}\}$ of ρth fractional domains as an example, our goal is to estimate the OT distance between $\boldsymbol{f}^{h,\rho}$ and $\boldsymbol{f}^{\ell,\rho}$ and transform it as the representation weight of this domain. To observe the feature distribution of each modality in the ρth fractional domain, a mini-batch of hyperspectral and LiDAR embeddings is taken to estimate the OT distance. Formally, defined the hyperspectral and LiDAR feature bags as $\boldsymbol{F}^{h,\rho} = [\boldsymbol{f}_1^{h,\rho}, \boldsymbol{f}_2^{h,\rho}, \ldots, \boldsymbol{f}_{n_b}^{h,\rho}]$ and $\boldsymbol{F}^{\ell,\rho} = [\boldsymbol{f}_1^{\ell,\rho}, \boldsymbol{f}_2^{\ell,\rho}, \ldots, \boldsymbol{f}_{n_b}^{\ell,\rho}]$, respectively, and the n_b is the batch size. To calculate the OT distance of feature embeddings, we adopt the discrete formal of OT. Suppose that the $\boldsymbol{\mu}^\rho = \sum_{i=1}^{n_b} \alpha_i \delta_{\boldsymbol{f}_i^{h,\rho}}$ and $\boldsymbol{\nu}^\rho = \sum_{i=1}^{n_b} \beta_i \delta_{\boldsymbol{f}_i^{\ell,\rho}}$, where $\delta_{\boldsymbol{f}_i^{h,\rho}}$ represent the Dirac function centered on $\boldsymbol{f}_i^{h,\rho}$, and the $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{n_b})$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_{n_b})$ are the weight vectors. Then we compute the 1-Wasserstein distance with a cosine distance ground-cost by

$$\begin{aligned}\mathcal{D}_{\mathrm{OT}}^{\rho} &= \inf_{\boldsymbol{\gamma} \in \Pi(\boldsymbol{\mu}^\rho, \boldsymbol{\nu}^\rho)} \mathbb{E}_{\left(\boldsymbol{f}_i^{h,\rho}, \boldsymbol{f}_j^{\ell,\rho}\right) \sim \boldsymbol{\gamma}} \left[c\left(\boldsymbol{f}_i^{h,\rho}, \boldsymbol{f}_j^{\ell,\rho}\right) \right] \\ &= \min_{\mathbf{T} \in \Pi(\mathbf{u}, \mathbf{v})} \sum_{i=1}^{n_b} \sum_{j=1}^{m} \mathbf{T}_{ij} \cdot c\left(\boldsymbol{f}_i^{h,\rho}, \boldsymbol{f}_j^{\ell,\rho}\right) \end{aligned} \tag{7}$$

where $\Pi(\boldsymbol{\mu}^\rho, \boldsymbol{\nu}^\rho) = \{\boldsymbol{T} \in \mathbb{R}_+^{(B^o \times B^o)} \mid \boldsymbol{T}\mathbf{1}_B^o = \boldsymbol{\alpha}, \boldsymbol{T}^T\mathbf{1}_B^o = \boldsymbol{\beta}\}$ and $c(\cdot, \cdot)$ is the cost matrix.

The cosine distance is selected as the cost function in our framework, that is,

$$c\left(\boldsymbol{f}_i^{h,\rho}, \boldsymbol{f}_j^{\ell,\rho}\right) = 1 - \frac{\left(\boldsymbol{f}_i^{h,\rho}\right)^\top \boldsymbol{f}_j^{\ell,\rho}}{\left\|\boldsymbol{f}_i^{h,\rho}\right\|_2 \left\|\boldsymbol{f}_j^{\ell,\rho}\right\|_2}. \tag{8}$$

It is important to note that since we aim to estimate the OT distance between two sets of high-dimensional feature vectors, and it is well known that Euclidean distance is relatively weak when handling high-dimensional data [55]. Therefore, we opted for the theoretically more suitable cosine similarity as the ground-cost matrix. In addition, since our primary motivation is not to optimize the OT technique, the choice of this ground-cost metric is trivial and can be further explored to enhance the method's performance.

The OT distance between HSI and LiDAR data in each fractional domain can be calculated by using (7). The reciprocal form after normalization is taken as the guidance vector, denoted as $\boldsymbol{W}_{\mathrm{OT}} = [\mathcal{W}_{\mathrm{OT}}^{0.1}, \ldots, \mathcal{W}_{\mathrm{OT}}^{1.0}]$, which can be calculated by

$$\boldsymbol{W}_{\mathrm{OT}} = \mathrm{softmax}\left(\left[1/\mathcal{D}_{\mathrm{OT}}^{0.1}, \ldots, 1/\mathcal{D}_{\mathrm{OT}}^{1.0}\right]\right). \tag{9}$$

Then, the vector $\boldsymbol{D}_{\mathrm{OT}}$ is viewed as attention weights to guide the fusion of feature representations from multiple fractional domains. The $\boldsymbol{W}_{\mathrm{OT}}$ vector is used as weighting coefficients for the multimodal feature representations from different domains, that is,

$$\boldsymbol{f}^{\mathrm{OT}} = \sum_{\rho=0}^{1.0} \mathcal{W}_{\mathrm{OT}}^{\rho} \cdot \boldsymbol{f}^{m,\rho} \tag{10}$$

where $\boldsymbol{f}^{m,\rho} = \mathrm{Concat}(\boldsymbol{f}^{h,\rho}, \boldsymbol{f}^{\ell,\rho})$. Finally, a classification head is employed to project the optimal representation $\boldsymbol{f}^{\mathrm{OT}}$ into a C-dimensional vector, where C is the number of classes. During the training phase, cross-entropy loss is used to optimize the network. In the testing phase, the feature representation of test samples is obtained through the OT-guided fusion of features from multiple fractional domains.

In summary, the two key intuitive designs in our method will provide a certain level of assurance that DMFraF has advantages over existing spatial domain-based methods. First, the model's expressive capacity, based on a multifractional domain design, is superior to that of models relying solely on spatial domain features. In other words, the models based on a

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TABLE I
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE HOUSTON2013 DATASET

No.	Land cover type	Training Set	Testing Set
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
Total	-	2832	12197

TABLE II
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE MUUFL DATASET

No.	Land cover type	Training Set	Testing Set
1	Trees	1138	21981
2	Mostly grass	383	3845
3	Mixed ground surface	868	5918
4	Dirt and sand	230	1570
5	Road	859	5732
6	Water	43	418
7	Building shadow	266	1938
8	Building	1228	4876
9	Sidewalk	295	1057
10	Yellow curb	56	121
11	Cloth panels	113	143
Total	-	6088	47599

TABLE III
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE AUGSBURG DATASET

No.	Land cover type	Training Set	Testing Set
1	Forest	768	5295
2	Park	124	139
3	Residential	1718	41499
4	Industrial	513	11907
5	Farm	402	6840
6	Cemetery	205	1377
7	Allotments	455	3851
8	Meadow	592	622
9	Commercial	595	4139
10	Recreation Ground	317	2636
11	Retail	357	1285
12	Scrub	642	489
13	Grass	175	148
14	Heath	45	120
Total	-	7676	80347

multifractional domain can surpass the upper limit of representational capacity inherent in models that rely solely on spatial domain features. Second, we leverage OT to dynamically coordinate the contributions of multiple fractional domains, which allows for the effective integration of multimodal representations from different fractional domains and avoids the heuristic settings of fractional orders. Even when certain fractional domains do not contribute positively to the task, our framework will adaptively down-weights their influence.

III. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, three widely used HSI and LiDAR datasets are first introduced. Then, the implementation details and evaluation metrics are given. Next, the comparison results and corresponding analysis with various state-of-the-art methods are reported and discussed. Finally, extensive experiments, including related ablation, detailed experiments of FraFT, and parameters studies, are conducted and analyzed.

A. Datasets Description

Three hyperspectral and LiDAR benchmark datasets are selected for experiments, and they have been utilized for experimental evaluation by many related articles. To evaluate the classification performance of different methods more fairly, three datasets are all separated into disjoint training and testing sets. The training and testing set of the Houston2013 dataset have been presplit by the GRSS community. However, there are no standard separate training and testing sets for the MUUFL and Augsburg datasets, hence we processed the ground truth maps ourselves to obtain disjoint datasets. Specifically, we randomly selected a certain number of samples for each class and then set up randomly sized windows around each sample. We included samples within these windows of the same class in the training set. After obtaining the training set, we heuristically adjusted the class distributions to ensure that the number of training samples per class was as close as possible. However, due to the significant imbalance between rare and common classes in the MUUFL and AUB datasets, completely overcoming class imbalance in the training set remains highly challenging. As quantitatively shown in Tables I–III, a certain degree of disparity persists between rare and frequent classes in the MUUFL and Augsburg datasets despite these efforts. The details of these datasets are given as follows.

1) Houston2013 Dataset: This dataset was acquired by the ITRES CASI-1500 imaging sensor in June 2012, which comprises an HS image and LiDAR-derived digital surface model (DSM) data [56]. It covers the University of Houston campus and its surrounding urban area, Texas, USA, containing 349 $\times$ 1905 pixels with a spatial resolution of 2.5 m. The HSI of Houston2013 dataset consisting of 144 spectral bands covering 380–1050 nm region at 10-nm spectral resolution, and the co-registered LiDAR data containing only one channel that records height information. Its ground truth contains

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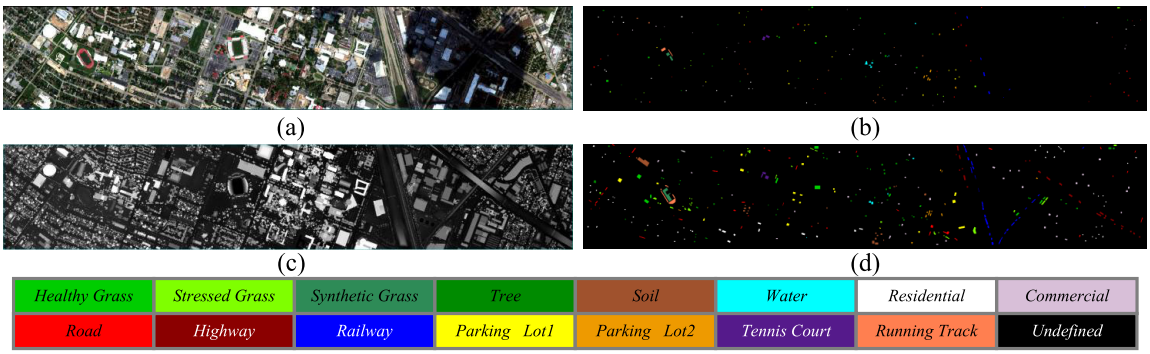


Fig. 5. Overview of the Houston2013 dataset. (a) False-color image for hyperspectral data. (b) Training set. (c) Grayscale image for LiDAR. (d) Testing set.

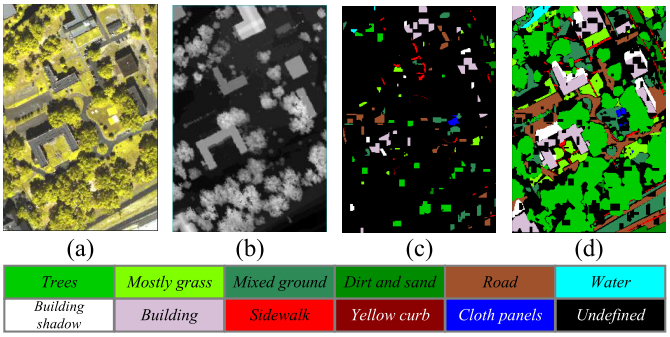


Fig. 6. Overview of the MUUFL dataset. (a) False-color image for hyperspectral data. (b) Grayscale image for LiDAR. (c) Training set. (d) Testing set.

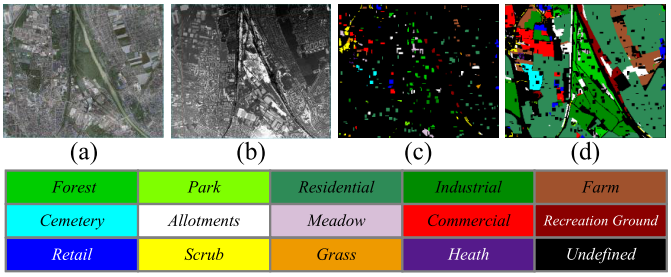


Fig. 7. Overview of the Augsburg dataset. (a) False-color image for hyperspectral data. (b) Grayscale image for LiDAR. (c) Training set. (d) Testing set.

15 029 pixel-level labels of 15 meaningful classes, and the standard training and testing sets given by the 2013 GRSS Data Fusion Contest[1] are adopted. The false-color and ground truth maps of the Houston13 dataset are shown in Fig. 5, and the detailed sample sizes of different classes are listed in Table I.

2) MUUFL Dataset: The MUUFL dataset consists of HSI data captured by the ITRES CASI-1500 sensor and LIDAR-based DSM data collected by the Gemini ALTM LiDAR sensor [57]. It was acquired in November 2010 and covers the University of Southern Mississippi Gulfport Campus, Long Beach, Mississippi, USA. The HSI data have 64 available spectral channels ranging from 375 to 1050 nm and a spatial size of 325 × 220 pixels. The LiDAR modality contains across track with 0.54-m spatial resolution and along track with 1.0-m spatial resolution. As shown in Fig. 6, to avoid the overlap of training and test data, the dataset is split into disjoint training and test sets like the above-mentioned Houston2013 dataset. There are 53 687 labeled samples, and the detailed training and testing c was summarized in Table II.

3) Augsburg Dataset: The third dataset comprises an EeteS-simulated EnMAP HSI and a 3K DSM, which comes as a part of a multimodal benchmark MDAS provided in [58]. This subarea covers a subregion of 10 km^2 of the city of Augsburg, Germany, containing 14 classes generated by manual labeling. Fig. 7 presents the false-color and ground truth maps of the Augsburg dataset. It can be seen that the HSI data and the LiDAR data were spatially adjusted to a 10 m GSD, resulting in a total of 300 × 360 pixels. Similarly, the training and testing sets were separated into disjoint forms and reported in Table III.

B. Implementation and Evaluation Metrics

All experiments are implemented on the same hardware platform with GTX-3090 GPU, Intel 4210R CPU, and 45G memory. Concretely, the Adam is adopted as the optimizer with a learning rate of $1e^{-3}$, and the weight decay is set to $1e^{-4}$ for the stable training of the network. In addition, the training epochs are empirically set to 100 for fully finding a converged status with satisfying results. In terms of the settings of DMFraF's specific parameters, the depth and embedding dimensions of the Transformer branch are set as 4 and 192, respectively. Notably, the LiDAR modality in MUUFL consists of two spectral bands. In our method, we directly feed the raw LiDAR data into the dedicated LiDAR mapping layer without implementing any preprocessing or feature engineering. The overall accuracy (OA), class-specific accuracy (AA), and the Kappa coefficient are employed to evaluate the classification performance. To avoid biased estimates, all experiments were conducted with ten independent tests.

C. Comparison Results and Analysis

To evaluate the comprehensive performance of the DMFraF, a list of advanced HSIC methods are implemented on the same experimental platform, including three conventional classification methods and four joint classification methods. In terms of the conventional HSIC methods, three representative methods

[1]http://www.classic.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

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TABLE IV
CLASSIFICATION ACCURACIES OF METHODS ON THE HOUSTON2013 DATASET

Class	SSFTT [17]	HyperSIGMA [21]	DSFormer [59]	CoupledCNN [30]	ExViT [41]	MFT [42]	FrIT [49]	UACL [44]	CALC [60]	DCMNet [61]	DMFraF
1	77.99±0.82	81.06±0.86	81.75±0.67	83.07±0.06	83.34±2.77	80.70±0.60	**100.00±0.00**	82.60±0.67	80.72±0.78	82.81±0.33	82.96±0.18
2	83.23±0.88	84.21±0.44	84.61±0.50	84.92±0.28	84.20±1.19	81.17±2.72	**97.74±4.12**	85.13±0.04	84.32±0.43	85.15±0.08	84.93±0.47
3	91.13±2.78	97.35±1.31	82.50±2.55	95.37±1.94	95.66±4.71	96.87±4.98	97.92±4.29	99.96±0.08	91.73±1.07	97.23±0.82	**100.00±0.00**
4	86.93±1.07	90.19±0.87	91.76±0.66	93.85±2.06	92.71±1.92	92.18±2.33	**99.70±0.48**	95.83±2.93	91.17±3.40	98.11±0.38	97.65±1.04
5	98.31±0.58	99.74±0.48	99.98±0.04	99.80±0.31	99.61±0.36	98.24±1.86	99.93±0.02	99.98±0.04	99.98±0.02	99.91±0.20	**99.98±0.04**
6	93.15±3.71	96.92±1.14	90.35±1.62	97.90±2.54	94.62±4.74	92.59±4.00	97.02±4.94	97.76±1.62	97.06±1.88	99.16±1.68	**99.30±0.77**
7	77.95±2.27	84.80±1.73	83.64±2.67	92.60±2.40	88.78±4.49	84.44±4.77	**96.46±1.11**	86.03±3.31	87.91±1.53	79.76±3.06	88.27±1.51
8	68.98±3.05	75.88±3.53	83.10±5.97	87.09±4.62	89.44±5.64	74.95±8.85	77.62±1.25	82.81±1.73	82.55±3.55	85.57±3.39	**89.50±0.70**
9	83.15±2.76	87.72±1.38	83.68±2.02	86.29±3.72	86.77±2.91	81.62±3.70	**93.42±1.70**	90.07±1.29	89.60±3.42	92.64±2.28	92.88±1.73
10	62.72±3.13	54.65±2.61	59.04±3.50	64.44±2.09	65.78±1.94	57.07±7.33	67.00±8.80	70.02±7.81	66.21±2.83	80.60±7.36	**80.95±0.70**
11	86.81±3.69	74.42±1.54	91.65±3.12	93.74±2.88	87.31±6.30	93.06±2.25	84.30±5.73	98.44±0.93	96.57±5.09	98.10±1.30	**99.11±0.42**
12	96.73±0.79	83.17±2.64	94.81±1.39	90.31±6.40	90.62±3.66	93.31±2.11	93.93±6.69	94.58±1.27	95.33±0.59	**99.81±0.50**	96.10±1.35
13	86.46±2.69	91.02±0.76	84.21±2.32	90.67±2.87	88.67±3.22	81.54±4.94	92.47±4.13	88.70±3.24	87.51±3.04	87.37±1.60	**94.67±1.74**
14	100.00±0.00	100.00±0.00	97.41±1.13	98.26±2.06	97.25±2.60	99.19±1.62	99.42±1.49	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
15	89.60±7.56	98.82±1.66	94.50±3.63	95.96±5.67	95.92±6.89	98.22±1.57	99.52±0.35	99.58±0.52	98.14±1.78	100.00±0.00	**100.00±0.00**
OA(%)	83.53±0.61	83.73±0.59	85.94±1.31	88.70±0.47	87.96±1.23	85.18±0.82	91.03±0.65	89.81±0.52	88.42±0.38	91.15±0.83	**92.29±0.18**
AA(%)	85.54±0.61	86.66±0.49	86.87±1.26	90.28±0.52	89.38±1.24	87.01±0.81	93.09±0.51	91.43±0.57	89.92±0.26	92.47±0.69	**93.75±0.11**
Kappa	0.82±0.01	0.82±0.01	0.85±0.01	0.88±0.01	0.87±0.01	0.84±0.01	0.90±0.00	0.89±0.01	0.87±0.01	0.90±0.01	**0.92±0.00**

TABLE V
CLASSIFICATION ACCURACIES OF METHODS ON THE MUUFL DATASET

Class	SSFTT [17]	HyperSIGMA [21]	DSFormer [59]	CoupledCNN [30]	ExViT [41]	MFT [42]	FrIT [49]	UACL [44]	CALC [60]	DCMNet [61]	DMFraF
1	91.15±0.67	89.21±1.25	91.42±0.38	92.73±1.57	92.60±0.83	89.18±1.12	**96.47±0.40**	95.62±0.21	83.64±0.74	92.94±0.24	93.54±0.69
2	72.84±1.42	76.94±3.37	74.58±0.97	77.78±6.09	75.04±5.52	69.45±2.48	71.27±3.03	70.21±3.50	72.85±5.21	70.10±2.44	**82.63±2.53**
3	73.81±0.63	75.67±1.00	78.36±0.71	71.81±12.94	80.78±3.66	73.36±2.10	79.93±2.74	**87.25±3.52**	74.16±4.15	79.94±1.76	80.33±2.29
4	60.51±6.13	66.05±6.40	66.96±5.06	80.87±8.63	78.98±8.70	73.63±6.57	**84.44±3.21**	58.89±10.94	73.40±8.59	64.74±0.99	78.27±5.22
5	87.74±0.77	87.09±2.32	84.98±0.61	90.87±3.19	90.41±1.10	85.84±1.91	81.62±2.71	93.27±0.48	91.07±2.02	89.72±1.05	**93.80±0.72**
6	91.44±2.62	70.29±8.10	71.53±8.39	76.41±17.02	96.89±3.91	98.76±1.11	84.15±2.29	74.26±19.11	93.78±5.70	**97.80±1.77**	91.05±4.34
7	61.97±2.60	69.70±1.19	71.54±1.29	76.73±5.98	**81.93±2.72**	65.49±2.44	79.12±2.69	70.38±6.86	62.88±3.83	64.50±3.74	77.80±5.18
8	93.59±1.07	91.17±1.58	93.10±0.47	**96.05±1.37**	94.11±1.31	92.11±0.69	91.37±1.41	96.60±0.52	94.56±1.14	94.52±0.68	95.94±0.53
9	46.43±2.12	54.99±4.86	56.58±3.82	60.47±6.42	64.97±5.31	42.37±3.75	40.49±2.76	71.07±5.09	36.25±6.27	57.79±1.50	**75.95±4.20**
10	19.50±4.86	29.75±7.15	19.01±1.73	39.67±7.75	42.48±4.41	23.14±5.33	8.71±1.97	61.98±14.59	6.28±14.04	21.32±5.61	**70.74±8.37**
11	55.25±17.76	66.71±15.57	50.21±4.09	78.39±11.70	74.69±11.79	55.66±20.26	68.13±2.65	65.87±16.28	58.04±4.61	75.66±6.77	91.75±3.12
OA(%)	83.88±0.38	83.78±1.42	84.96±0.26	86.96±1.37	87.96±0.51	82.82±0.50	85.73±0.41	89.32±0.53	85.80±0.55	86.06±0.28	**89.67±0.36**
AA(%)	68.57±1.37	70.69±2.85	68.93±1.47	76.53±1.91	79.35±1.71	69.91±1.72	71.43±0.34	76.95±3.15	68.81±2.97	73.00±0.88	**84.71±0.95**
Kappa	0.78±0.00	0.78±0.02	0.80±0.00	0.83±0.01	0.84±0.0.01	0.77±0.01	0.81±0.00	0.85±0.01	0.81±0.01	0.81±0.00	**0.86±0.00**

TABLE VI
CLASSIFICATION ACCURACIES OF METHODS ON THE AUGSBURG DATASET

Class	SSFTT [17]	HyperSIGMA [21]	DSFormer [59]	CoupledCNN [30]	ExViT [41]	MFT [42]	FrIT [49]	UACL [44]	CALC [60]	DCMNet [61]	DMFraF
1	85.31±1.05	83.25±0.39	**86.23±2.37**	84.96±3.81	84.57±1.20	82.45±1.22	75.32±3.22	80.52±2.09	85.45±3.46	85.68±0.47	81.92±0.84
2	71.66±2.97	70.94±4.23	34.39±5.03	60.36±8.41	72.23±3.61	74.25±6.09	12.24±3.80	64.21±17.06	58.56±3.75	65.47±2.61	**75.76±3.71**
3	89.22±1.27	87.52±0.53	87.03±1.09	84.39±4.61	87.87±1.02	85.96±0.61	**93.96±0.67**	90.78±2.39	92.64±1.50	91.77±0.52	90.14±0.35
4	64.35±1.41	66.45±2.01	56.08±2.67	58.79±8.69	64.17±2.21	59.66±2.05	**85.70±0.61**	48.75±17.12	68.48±2.44	61.38±1.16	72.46±1.17
5	73.54±2.49	69.40±2.61	67.09±2.07	69.81±7.55	72.35±2.25	67.84±4.61	**90.90±1.23**	66.24±8.55	75.60±1.43	75.68±1.29	78.80±1.34
6	86.41±2.91	76.30±3.76	76.46±1.80	77.28±4.74	84.92±3.06	82.61±4.60	76.72±3.15	79.34±8.77	83.27±4.54	91.75±2.36	**94.41±1.19**
7	69.68±2.86	70.43±1.35	65.90±1.91	**72.64±8.32**	71.86±2.22	64.45±3.59	69.24±1.97	48.63±11.01	62.15±6.81	69.72±1.36	68.24±0.89
8	76.75±0.87	76.40±1.61	74.86±2.69	76.66±7.00	77.85±1.25	73.31±1.09	26.04±2.63	74.92±6.13	73.60±2.10	76.05±1.70	**78.12±0.83**
9	69.03±2.41	68.38±1.02	66.77±3.05	61.65±16.22	71.74±2.42	64.17±1.55	50.90±1.55	**82.56±1.31**	73.04±5.03	76.15±1.74	79.01±1.92
10	**88.43±1.39**	82.38±1.56	76.51±1.98	70.97±11.08	81.98±2.22	82.85±1.97	74.46±0.98	81.28±7.48	83.38±5.64	81.23±1.24	85.74±1.11
11	65.59±3.50	62.52±4.25	46.55±1.45	50.35±13.65	61.48±4.37	**69.31±3.15**	51.20±3.28	48.75±22.04	55.22±6.87	60.78±5.48	63.89±3.42
12	74.19±1.66	71.70±1.46	70.76±0.86	70.65±8.58	73.21±2.04	65.28±2.64	21.44±3.49	**77.97±5.92**	67.16±12.63	76.36±0.96	76.95±1.85
13	31.62±3.15	29.32±1.94	20.68±1.01	28.65±5.36	31.55±2.79	29.60±3.69	10.12±2.41	29.22±6.90	21.08±12.13	**33.92±1.56**	31.69±4.04
14	76.83±2.66	74.17±1.05	50.83±3.21	74.25±13.03	76.00±3.87	76.00±3.74	29.03±3.21	77.50±8.60	45.32±14.22	81.00±1.43	**88.25±3.08**
OA(%)	81.17±0.72	79.67±0.38	77.00±0.56	76.25±2.73	80.24±0.54	77.35±0.70	80.33±0.67	77.78±3.74	83.04±0.45	82.41±0.19	**83.59±0.28**
AA(%)	73.04±0.28	70.65±0.83	62.87±0.46	67.24±1.49	72.27±0.50	69.84±1.28	54.80±0.70	67.90±1.00	67.50±3.25	73.55±0.62	**76.10±0.40**
Kappa	0.74±0.01	0.72±0.01	0.68±0.01	0.67±0.03	0.73±0.01	0.69±0.01	0.72±0.00	0.69±0.04	0.76±0.01	0.75±0.00	**0.77±0.00**

are selected for comparison, including spectral–spatial feature tokenization Transformer (SSFTT [17]), hyperspectral intelligence comprehension foundation model (HyperSIGMA [21]), and a dual selective fusion Transformer (DSFormer) network [59]. With respect to the HSI-LiDar joint classification algorithms, we mainly select CNN-based (CoupledCNN [30] and UACL [44]), Transformer-based (ExViT [41], MFT [42], and FrIT [49]) multimodal joint feature learning models, a coupled adversarial learning based joint classification method (CALC [60]) and a dynamic cross-modal feature interaction network (DCMNet [61]). It is noteworthy that FrIT is also a hyperspectral joint classification model based on FrFT. All the methods are implemented following the parameter-configurations provided in corresponding researches. In addition, for the conventional HSIC methods, in order to avoid violating the inherent assumptions of the models by introducing LiDAR data, only the HSI modality was retained as the model input.

Tables IV–VI present the quantitative experimental results of all methods on the Houston2013, MUUFL, and Augsburg datasets, respectively. It can be observed that our DMFraF achieves the best joint classification performance on all three datasets and significantly outperforms the second-best method. As shown in Table IV, on the Houston2013 dataset, DMFraF surpasses the second-best method by 1.14% in OA, 0.66% in AA, and 0.20 in Kappa, with higher accuracies in the majority of categories. It is particularly noted that the tenth class in the Houston2013 dataset, where there is a significant distribution

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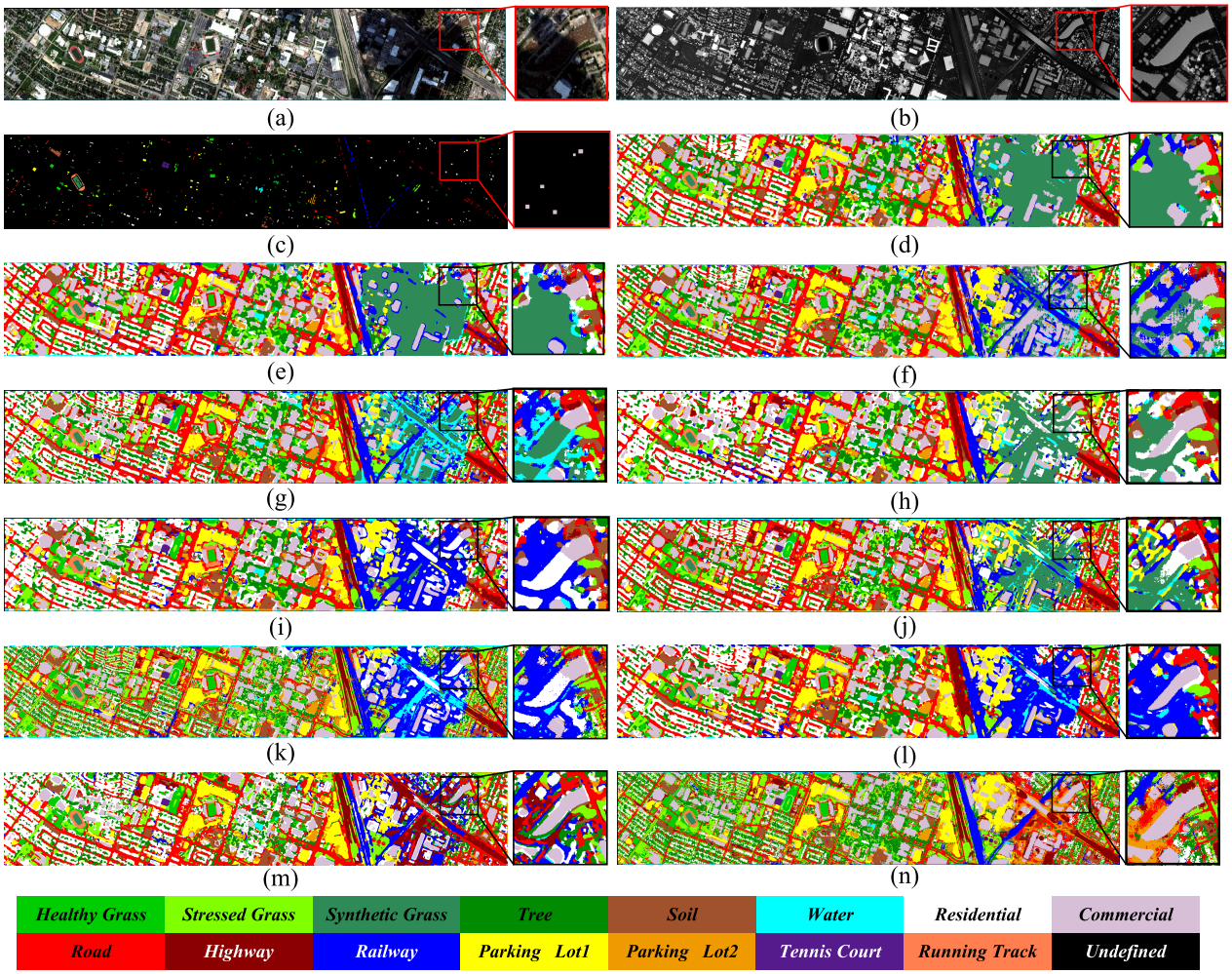


Fig. 8. Classification maps of different methods on Houston 2013 dataset. (a) Pseudo-color image for hyperspectral data. (b) Grayscale image for LiDAR. (c) Ground truth map. (d) SSFTT. (e) HyperSIGMA. (f) DSFormer. (g) CoupledCNN. (h) ExViT. (i) MFT. (j) FrIT. (k) UACL. (l) CALC. (m) DMFNet. (n) DMFraF.

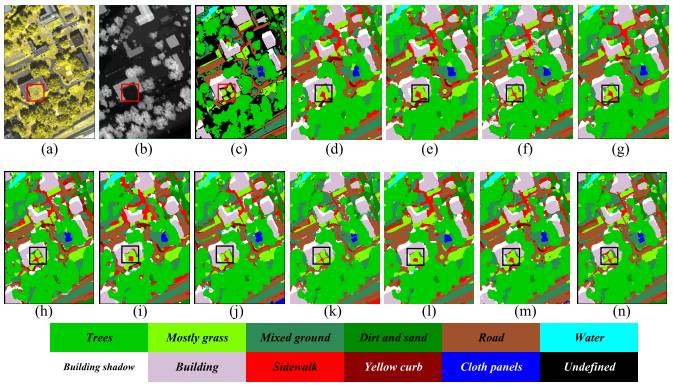


Fig. 9. Classification maps of different methods on MUUFL dataset. (a) Pseudo-color image for hyperspectral data. (b) Grayscale image for LiDAR. (c) Ground truth map. (d) SSFTT. (e) HyperSIGMA. (f) DSFormer. (g) CoupledCNN. (h) ExViT. (i) MFT. (j) FrIT. (k) UACL. (l) CALC. (m) DMFNet. (n) DMFraF.

difference between the test and training data. In the training set, the model primarily observes samples from the left region, while the test samples are mainly concentrated in the right region, which is further obstructed by cloud cover. Surprisingly, for this class, DMFraF exhibits a significantly excellent classification performance compared to other methods (with an accuracy exceeding that of the second-best FrIT by 13.95%), suggesting the superiority and better generalization of our approach.

For the MUUFL dataset, it can be observed from Table V that DMFraF surpasses the second-best method by 0.35% in OA, 5.36% in AA, and 0.1 in Kappa. Particularly, the advantage of DMFraF over other methods is significantly evident in terms of AA, as it performs well in each class. In contrast, other methods tend to overfit to certain classes due to highly imbalanced samples, resulting in underfitting for some classes. For instance, in the tenth class (i.e., "Road"), DMFraF outperforms other methods by large margins (i.e., exceeding the second-best method by 8.76%). This is because the number of samples in the tenth class is significantly fewer compared to other categories, and most of the counterparts are prone to fit categories with a larger proportion of samples. Moreover, UACL achieves an OA that is very close to that of our method on this dataset, primarily due to its strong performance on majority classes. However, its performance in terms of AA is significantly lower than that of our method, indicating that UACL tends to overfit to the majority classes to some extent. It is evident that DMFraF performs well across

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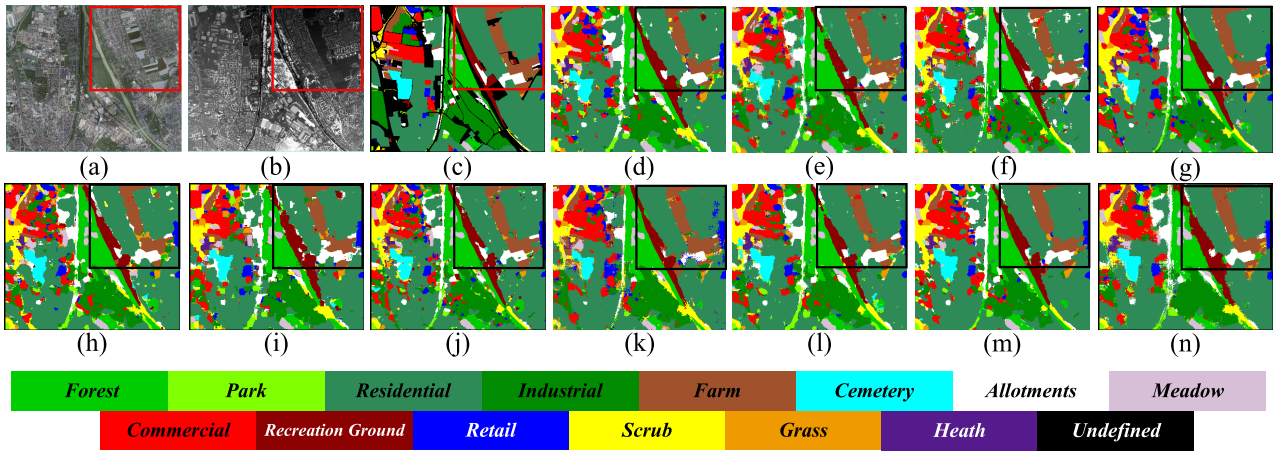


Fig. 10. Classification maps of different methods on AUB dataset. (a) Pseudo-color image for hyperspectral data. (b) Grayscale image for LiDAR. (c) Ground truth map. (d) SSFTT. (e) HyperSIGMA. (f) DSFormer. (g) CoupledCNN. (h) ExViT. (i) MFT. (j) FrIT. (k) UACL. (l) CALC. (m) DMFNet. (n) DMFraF.

all categories, verifying that the proposed method can alleviate the long-tail problem to some extent.

Table VI reports the experimental results of all methods on the Augsburg dataset. It can be observed that DMFraF surpasses other counterparts by at least 0.55% in OA, 2.55% in AA, and 0.1 in Kappa. It is noteworthy that, unlike the above-mentioned two datasets where joint classification methods significantly outperformed conventional HSIC methods, the performance of some joint classification methods on this dataset is even inferior to single-modal HSIC methods. This may be attributed to the complex distribution of LiDAR modality in this dataset, while the distribution of actual land covers is relatively simple. This complexity makes it challenging for multimodal methods to capture complementary patterns and may instead interfere with the prediction mechanism learned from the hyperspectral modality. Although the OA of SSRN is relatively remarkable, SSRN performs poorly on the AA metric. This is because SSRN performs well on categories with a large sample proportion (such as classes 1, 3, and 4) but performs poorly on categories with fewer samples (such as classes 13 and 14). As a result, the model achieves a high OA but a low AA. As expected, DMFraF still achieves the best classification performance. This is mainly attributed to multimodal representation from multiple fractional domains and the dynamic fusion mechanism of DMFraF. During the testing process, DMFraF can flexibly adjust the optimal multimodal representation of different samples from multiple fractional domains, thereby better coordinating complementary patterns between different modalities. Furthermore, except for the 13th class, the class-specific accuracies of DMFraF are excellent in other categories, yielding the superiority and robustness of our method.

Figs. 8–10 illustrate the qualitative classification results of all methods on three datasets. It can be observed from the figures that the classification maps of DMFraF have clearer boundaries and the land-covers distributions are closer to ground truth. Thanks to the C^2AT encoder which can reduce the interference of noisy pixels within the patch, the classification maps of DMFraF are not excessively smooth like other methods. Moreover, to explore the multimodal representation capability of DMFraF, we deliberately selected a challenging area (as shown in the rectangular areas of Fig. 8) in the Houston dataset for studying. In this rectangular area, part of the eighth class (i.e., "Commercia") in the HSI modality is partially obscured by cloud cover while another part is not, yet its LiDAR modality exhibits discriminative semantics. From the classification maps, it can be observed that, except for DMFNet, ExViT, and DMFraF, all other methods misclassify the pixels within the cloud cover, and even ExViT and DMFNet cannot completely classify the entire area correctly. Only DMFraF accurately classifies the overall region, validating the proposed method's effectiveness in capturing complementary patterns between modalities. Furthermore, DMFraF exhibits significantly clearer classification results in the cloud-covered regions of the Houston2013 dataset compared to other methods. Quantitative results suggest that our method not only has excellent performance in qualitative metrics but also yields superior quantitative performance and generalization ability in universal data.

D. Effectiveness Analysis of DMFraF

1) Ablation Study: To further explore the performance gains of the core parts of our method, i.e., C^2AT encoder, Multiple fractional Fourier domains, and OT-guided dynamic fusion, the ablation studies is designed for analysis of each components. Table VII presents the ablation results on the three datasets, where "✓" indicates the module being added, and "×" indicates its absence. The three crucial modules are sequentially added to the baseline, elucidating how each component contributes to the overall compositional model. Notably, the C^2AT encoder constitutes a separate branch (i.e., channel encoder), and it is entirely removed from the baseline. Moreover, as OT is proposed to dynamically fuse representations from multiple fractional domains, FrFT must be performed before adding OT.

A key observation is that the addition of each module leads to varying degrees of performance improvement in terms of OA, AA, and Kappa, underscoring the positive contribution of each module to the overall outcome. As expected, when

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TABLE VII
ABLATION STUDY OF DMFRAF ON THREE DATASETS

Variants	Modules			Results		
	C^2AT Encoder	Multiple fractional Fourier domains	Optimal transport	OA	AA	Kappa
Houston2013	×	×	×	90.81±0.44	92.18±0.41	0.90±0.00
	✓	×	×	91.38±0.36	92.90±0.31	0.91±0.00
	✓	✓	×	91.84±0.26	93.33±0.18	0.91±0.00
	✓	✓	✓	**92.29±0.18**	**93.75±0.11**	**0.92±0.00**
MUUFL	×	×	×	85.20±0.39	75.07±1.21	0.80±0.01
	✓	×	×	88.04±0.54	82.69±1.62	0.84±0.01
	✓	✓	×	89.45±0.26	82.33±0.66	0.86±0.00
	✓	✓	✓	**89.67±0.36**	**84.71±0.95**	**0.86±0.00**
Augsburg	×	×	×	80.51±0.27	72.23±0.20	0.73±0.00
	✓	×	×	81.92±0.25	74.75±0.40	0.75±0.00
	✓	✓	×	82.18±0.47	74.77±0.40	0.75±0.01
	✓	✓	✓	**83.59±0.28**	**76.10±0.40**	**0.77±0.00**

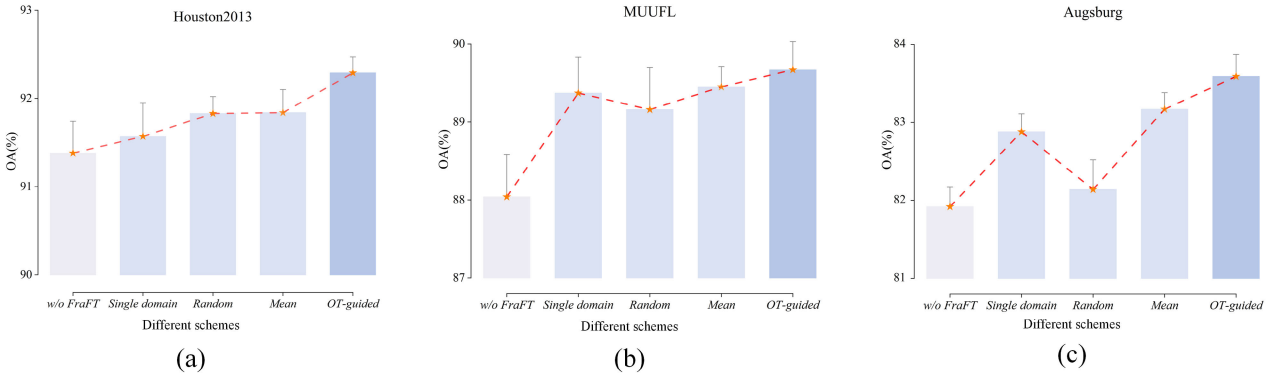


Fig. 11. Experimental exploration of the effectiveness of DMFraF on three datasets. (a) Houston2013. (b) MUUFL. (c) Augsburg.

TABLE VIII
ABLATION RESULTS OF DIFFERENT MULTIMODAL FUSION METHODS ON THREE DATASETS

	HSI data	LiDAR data	Concat	Sum	OT-DFraF	Houston2013	MUUFL	Augsburg
HSI-only	✓	×	×	×	×	87.13±0.58	88.28±0.58	81.70±0.15
LiDAR-only	×	✓	×	×	×	65.57±0.75	72.57±0.26	74.05±0.39
Fusion(concat)	✓	✓	✓	×	×	91.18±0.56	88.68±1.02	82.39±0.39
Fusion(sum)	✓	✓	×	✓	×	91.28±0.37	86.96±0.74	81.92±0.25
DMFraF	✓	✓	×	×	✓	**92.29±0.18**	**89.67±0.36**	**83.59±0.28**

all three components are added, our method achieves the best classification performance on three datasets. It is worth noting that the improvement in OA brought by OT on the MUUFL dataset is not as significant (i.e., 0.22%↑), but the improvement in the AA metric is remarkably significant (i.e., 1.84%↑). This is because the dynamic multimodal representation can debias the implicate data propensity to some extent, thereby reducing the long-tail effect.

2) Experimental Exploration on the Effectiveness of DMFraF: To further explore the effectiveness of the OT-guided dynamic fusion scheme, we extensively design multiple variants of DMFraF based on different various fractional domain selection methods. Specifically, five distinct selection strategies are chosen for comparative analysis as follows.

1) "*w/o FraFT*," a variant obtained via the removal of the FrFT and can be regarded as a baseline.
2) "Single domain," a variant with 0.5-order fractional domain, which encompasses equal spatial and frequency domain information.
3) "Random," a fractional domain selection strategy based on randomness, dynamically selecting a fractional domain for each input instance in a stochastic manner.
4) "Mean," a fusion strategy based on mean weighting, dynamically obtaining rich multimodal representations by averaging the feature vectors of multiple fractional domains.
5) "OT-guided," our proposed dynamic fusion scheme, is based on OT-guided methods to adjust the contributions of different fractional domains, thereby dynamically modeling optimal multimodal representations for each instance.

The joint classification performance of different variants on three datasets is illustrated in Fig. 11, where the mean and standard deviation of OA are employed as primary evaluation metrics. A fundamental observation is that the joint classification performance after incorporating fractional domain representations surpasses that of the baseline model, indicating the beneficial contribution of fractional domain features to the classification task. Moreover, the performance improves with

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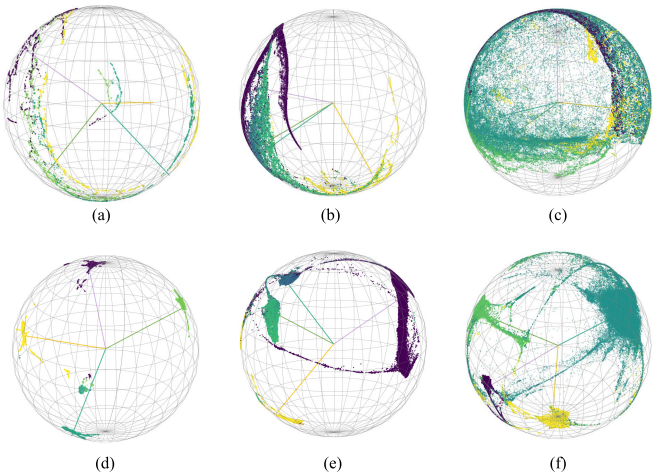


Fig. 12. (a)–(c) Projection of original features and (d)–(f) multimodal features of the four categories with the most severe interclass overlap on three datasets. (a) Original features of Houston2013. (b) Original features of MUUFL. (c) Original features of Augsburg. (d) Multimodal features of Houston2013. (e) Multimodal features of MUUFL. (f) Multimodal features of Augsburg. The scatter points on spheres are the feature embeddings, and the intersections of light dash-dot lines and sphere are the class features (mean vectors). Each color represents a class.

the increasing number of utilized fractional domains ("Mean" and "OT-guided" employ multiple fractional domains, while "Single domain" and "Random" only use one). This validates that fractional domains can enrich multimodal feature representations, thereby better acquiring task-discriminative feedback, and that a greater number of fractional domains can bring about richer information. Furthermore, it is evident that our proposed OT-guided fusion strategy exhibits significant advantages over other strategies across all three datasets. This validates that the multidomain dynamic adjustment strategy guided by OT is remarkably superior to the approaches based on the single fractional domain and even outperforms fixed-weighted multidomain aggregation (i.e., "Mean"). This is because the OT-guided fusion strategy can dynamically adjust the contributions of different fractional domains, thereby modeling the optimal representation for classification tasks across various input instances.

3) Ablation Study of Multimodal Joint Classification: The joint classification of hyperspectral and LiDAR data aims to integrate complementary patterns between different modalities, thereby leveraging the advantages of both modalities for land cover recognition. To validate the multimodal fusion performance of our approach, we compared the classification results based on single-modality data (i.e., "HSI-only" and "LiDAR-only") and joint multimodal data on three datasets. Moreover, we also compared two simple yet effective nondynamic fusion methods, i.e., "Concat" and "Sum," where the contributions of the two modalities were set to be equivalent.

The classification results of different methods on the three datasets are summarized in Table VIII, where OA is used as the evaluation metric. As expected, methods based on multimodal joint classification generally outperform those based on a single modality. Our proposed DMFraF achieved the best classification accuracy on all three datasets (exceeding the second-best counterpart by 1.01%, 1.39%, and 1.20% on the Houston, MUUFL, and Augsburg datasets, respectively). It is worth noting that the advantage of the joint classification models based on concat and SUM compared to the HSI-only model is minimal on the MUUFL dataset. In fact, the result of SUM on MUUFL is even lower than that of "HSI-only." This is because they employ nondynamic fusion methods and assign equal contributions to the features of the two modalities in the multimodal representation, leading to negative fusion from redundant modality information in the joint classification task.

E. Visualization of Different Feature Spaces

The Umap technique is effective in reducing the dimensions of high-dimensional features while preserving their manifold structure in the high-dimensional feature space. To intuitively explore the effectiveness of multimodal feature representations, we employ Umap to visualize the original HSI data and multimodal representations of the categories with the most severe overlap (categories 7, 9, 10, and 11 in the Houston dataset; categories 1, 2, 3, and 4 in the MUUFL dataset; categories 1, 3, 4, and 5 in the Augsburg dataset) in a reduced dimensionality on a spherical coordinate system. For comparison, we also visualize the original features in the same manner.

Fig. 12 displays the visualization results of different feature embeddings on the three datasets. The scatter points on the sphere represent different feature projections, while solid lines and intersections with the sphere represent class features (centroid vectors). The first row of Fig. 12 illustrates the high-dimensional geometric structure of the original HSI modality, while the second row showcases the multimodal representations modeled by DMFraF. It is evident that the clusters of original features are scattered and different categories are overlapped together, making it difficult for linear classifiers to distinguish between different categories. In turn, the multimodal feature vectors obtained by DMFraF have more compact intraclass distance and exhibit larger interclass separability, intuitively demonstrating the effectiveness of DMFraF.

F. Parameter Analysis

1) Impact of Patch Size: Due to the inherent data characteristics of HSI, HSIC models typically assume that HSIs are spatially sparse while spectrally correlated. Therefore, to fully leverage spatial information, most existing deep models adopt patch-based hyperspectral cubes as inputs. Thanks to the outstanding performance of such types of models, this assumption is also employed in joint classification models based on HSI and LiDAR data. Particularly, larger patches contain richer spatial information but also introduce more noisy pixels. Conversely, smaller patches contain purer pixels but have relatively limited spatial information. Therefore, patch size is a crucial hyperparameter for joint classification models.

To explore the influence of this parameter on DMFraF, we compared the classification performance of DMFraF under different patch sizes on three datasets. Fig. 13 reports the OAs of DMFraF under different settings of patch sizes ranging from [7:2:21]. As shown in Fig. 13, it can be observed that

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TABLE IX
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON THREE DATASETS

	Method	SSFTT [17]	HyperSIGMA [21]	DSFormer [59]	CoupledCNN [30]	ExViT [41]	MFT [42]	FrIT [49]	UACL [44]	CALC [60]	DCMNet [61]	**DMFraF**
Houston2013	Training time(s)	1.02	6.99	3.03	0.93	2.24	1.26	**34.27**	5.20	4.75	1.85	3.14
	Testing time(s)	45.33	328.63	219.34	46.21	140.43	56.04	**445.94**	81.31	88.33	90.95	327.44
	FLOPs(M)	57.37	3119.21	16.77	9.79	45.84	35.02	**3371.98**	9.88	27.71	35.77	172.64
MUUFL	Training time(s)	1.29	11.54	6.52	1.35	4.15	1.85	**64.72**	10.71	4.38	3.83	5.76
	Testing time(s)	2.89	35.03	22.25	2.93	15.29	3.80	**41.65**	6.36	9.29	10.39	34.37
	FLOPs(M)	25.11	3110.47	9.08	5.94	45.51	15.11	**3206.65**	6.03	18.57	35.77	168.11
Augsburg	Training time(s)	4.37	15.82	9.07	3.76	8.26	3.80	**120.50**	20.96	10.89	5.34	9.55
	Testing time(s)	10.98	56.84	40.65	11.49	27.08	15.64	**89.31**	14.43	22.17	15.46	62.65
	FLOPs(M)	96.89	3130.05	26.31	14.56	45.84	59.53	**3574.50**	14.65	39.06	35.77	178.26

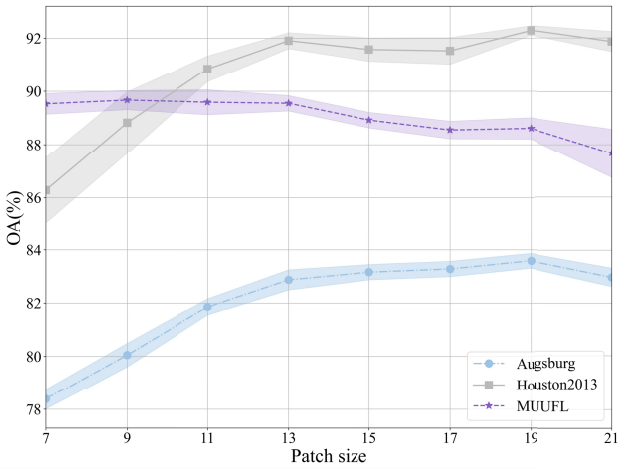


Fig. 13. Performance of DMFraF with different patch sizes on three datasets.

DMFraF achieves the best classification performance on the Houston and MUUFL datasets when the patch size is set to 19. However, the optimal patch size for the MUUFL dataset is 9. This may be attributed to the fact that the MUUFL dataset has a more detailed distribution of land covers compared to the other two datasets, thus requiring purer spatial information.

2) Computational Complexity: To comprehensively evaluate DMFraF, we further compared the computational cost among different deep learning-based methods. Table IX reports the computational complexity of different methods, which include the training times required for training one epoch, the testing times, and the floating point operations (FLOPs) on three datasets. In addition, we set the patch size and batch size to the same values for different methods (13 and 128, respectively) for fair comparisons. From the results in Table IX, both FrIT and HyperSIGMA exhibit longer runtime durations, which primarily stems from their substantially higher parameter counts. This computational characteristic is further corroborated by the FLOPs metric, where FrIT and HyperSIGMA suggest FLOPs values significantly exceeding those of other methods. Although CoupledCNN has the shortest runtime, its classification performance on the three datasets is unsatisfactory. It should be noted that DCMNet maintains nearly identical FLOPs across all three datasets. This phenomenon occurs because DCMNet requires hyperspectral input to be preprocessed via principal component analysis (PCA) for dimensionality reduction to a fixed spectral dimension. Consequently, despite the MUUFL dataset's LiDAR modality containing two distinct bands, the unified FLOPs measurements are also similar to other datasets.

Since the patch size and batch size are fixed for all methods, the runtime of one epoch is proportional to the computational cost. As shown in the Table IX, our method does not exhibit the shortest runtime. The computational cost of DMFraF primarily stems from two factors: first, the multiple FrFTs, and second, the computation of the OT distance. These additional data transformations will inevitably introduce additional computational overhead. It is noteworthy that while our method exhibits marginally higher FLOPs compared to some baseline approaches, it yields superior training efficiency as evidenced by the runtime metrics in Table IX. This performance advantage is primarily attributed to the parallel computing architecture embedded in our algorithm. Overall, our method, while not having the shortest runtime, exhibits acceptable computational cost, indicating a reasonable tradeoff between classification performance and computational overhead.

IV. CONCLUSION

In this article, a novel DMFraF is proposed for is proposed for hyperspectral and LiDAR data classification. Multiple fractional Fourier domains with different spatial and frequency components are incorporated into multimodal fusion to comprehensively capture the complementary patterns between HSI and LiDAR data from distinct views of different fractional domains. A dynamic fusion scheme guided by the OT technique is proposed to model the optimal multimodal representation for different input instances from multiple FraFT domains, which are capable of dynamically adjusting the contributions of fusion representations from different fractional domains. To extract purer modality-specific features, a channel aggregation block based on C^2AT encoder is proposed. The C^2AT encoder can effectively aggregate channel-wise features of central pixels into the spatial branch and compresses interference from noisy surroundings. The experimental results on three hyperspectral and LiDAR datasets yield that the proposed DMFraF outperforms seven collaborative classification methods, and ablation studies verify the effectiveness of three proposed constituent modules. DMFraF primarily focuses on incorporating fractional domain information into the joint classification task of hyperspectral and LiDAR data, without placing significant emphasis on the information imbalance between modalities. In future work, we plan to design more effective modality projection strategies to better compensate for the informational disparity caused by intermodal imbalance.

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